Fluids & Hydrostatics: A Mini-Lecture

Ellie Su

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1 Pressure: Definition & Basics

- **Pressure** is force per area:

$$P = \frac{F}{A}, \qquad [P] = \text{Pa} = \text{N/m}^2.$$

- At sea level, $P_{\text{atm}} \approx 1.013 \times 10^5$ Pa (1 atm).
- In a **static fluid**, the contact force on a wall is *normal* to the surface. If shear appears, the fluid will flow.
- **Gauge vs. absolute:** $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$.
- **Compressibility:** For our purposes, *liquids are nearly incompressible* (ρ taken constant), while *gases are compressible* (density can vary noticeably with P, T).

2 Hydrostatic Pressure

Let z be vertical, positive *upwards*. A thin horizontal slab (area A, thickness Δz) in a static liquid of density ρ has forces: bottom pressure $P(z)\,A$ upward, top pressure $P(z+\Delta z)A$ downward, and weight $\rho g A \Delta z$ downward. Static equilibrium:

$$P(z)A - P(z+\Delta z)A - \rho g A \Delta z = 0 \quad \Longrightarrow \quad \frac{dP}{dz} = -\rho g.$$

Integrating from a reference level z_0 where $P = P_0$ gives

$$\boxed{P(z) = P_0 + \rho g\,(z_0 - z) = P_0 + \rho g h,}$$

with $h = z_0 - z$ the depth below the reference. A uniform overpressure ΔP in a sealed container simply adds everywhere:

$$P(z) = (P_0 + \Delta P) + \rho g h.$$

Rules of thumb. $\sim 10\,\text{m}$ of water $\approx 1\,\text{atm}$ extra; $1\,\text{cm}$ of water $\approx 98\,\text{Pa}$.

3 Pascal's Principle

> *A pressure change applied to a confined fluid is transmitted undiminished to every point of the fluid and the walls.*

For two pistons connected to the same reservoir:

$$\Delta P = \frac{F_1}{A_1} = \frac{F_2}{A_2} \;\Rightarrow\; \boxed{F_2 = F_1\,\frac{A_2}{A_1}}.$$

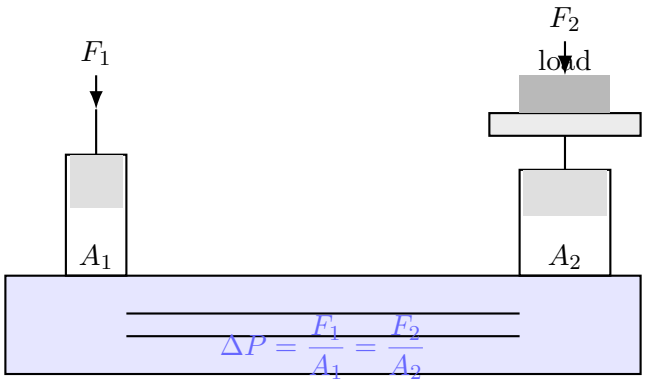


Figure 1: Two-cylinder hydraulic press: a small input piston drives pressure through a pipe to a larger output piston with a load platform.

4 Continuity & Bernoulli

For steady, incompressible flow:

$$\boxed{A_1 v_1 = A_2 v_2} \quad \text{(continuity)}$$

and along a streamline for *steady, incompressible, inviscid* flow with no pumps/turbines between the points:

$$\boxed{P + \tfrac{1}{2}\rho v^2 + \rho g z = \text{constant along a streamline.}}$$

Notes: Faster flow ⇒ lower static pressure (if height is the same). Do not compare pressures across unrelated streamlines.

5 Buoyancy & Floating

Archimedes: $F_b = \rho_{\text{fluid}} g\, V_{\text{disp}}$. For a floating object of volume V_o and density ρ_o:

$$\rho_{\text{fluid}} g\, V_{\text{disp}} = \rho_o g\, V_o \;\Rightarrow\; \frac{V_{\text{disp}}}{V_o} = \frac{\rho_o}{\rho_{\text{fluid}}}.$$

(*Iceberg:* $\rho_{\text{ice}} \approx 0.92\,\rho_{\text{water}}$ ⇒ about 92% submerged.)

Gravity matters: Without g there is no hydrostatic gradient and therefore no buoyancy (aside from effective "pseudo-buoyancy" in accelerated frames).

6 Concept Checks & Short Problems

CC1 — Straw and "Suction"

You drink through a vertical straw from a glass at sea level. Explain why the liquid rises in the straw when you inhale, using pressure language. What sets the maximum theoretical height for water?

CC2 — Snorkeling vs. Scuba

A swimmer breathes through a snorkel with her mouth at the surface while her chest is 1.2 m below the surface. Explain why inhalation is difficult. Why does scuba (air supplied at ambient pressure) remove this problem?

CC3 — Rock in a Boat

You sit in a big tank of water in a small boat holding a heavy rock. You throw the rock overboard and it sinks. Does the water level in the tank rise, fall, or stay the same?

CC4 — Ping-Pong Ball in a Funnel

A ping-pong ball sits in a funnel connected to a hose. When air is blown downward through the funnel, the ball remains seated. Explain qualitatively (using pressure distribution and flow attachment) why the ball stays in place, and why small tilts do not dislodge it.

7 Challenge Problem — Soap Bubble Pressure Jump

A soap bubble is a static, thin spherical shell of soap. The surface has tension γ (energy per area to create interface). **Question:** What is the pressure difference $\Delta P = P_{\text{in}} - P_{\text{out}}$ between the inside and outside of a spherical bubble of radius r in terms of r and γ? Use either a force balance on a hemispherical cut or an energy/work argument. State assumptions (thin film, uniform γ, equilibrium, negligible hydrostatic variation across the bubble).

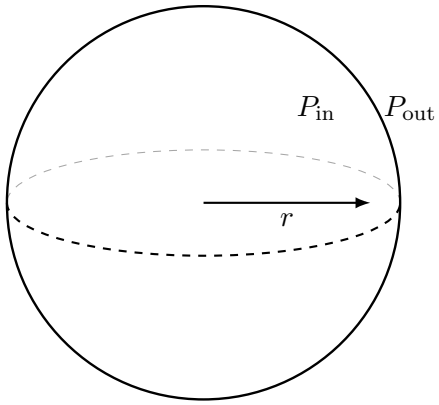


Figure 2: Spherical soap bubble for the pressure-jump derivation (challenge problem).

8 Additional Problems

Problem 1

A container of water is sitting on a scale. Originally, the scale reads $M_1 = 45$ kg. A block of wood is suspended from a second scale; originally the scale reads $M_2 = 12$ kg. The density of wood is 0.60 g/cm^3; the density of water is 1.00 g/cm^3. The block of wood is lowered into the water until *half* of the block is beneath the surface. What is the resulting reading on the scales?

1. $M_1 = 45$ kg and $M_2 = 2$ kg.
2. $M_1 = 45$ kg and $M_2 = 6$ kg.
3. $M_1 = 45$ kg and $M_2 = 10$ kg.
4. $M_1 = 55$ kg and $M_2 = 6$ kg.
5. $M_1 = 55$ kg and $M_2 = 2$ kg.

Problem 2

A 3.0 cm thick layer of oil with density $\rho_o = 800$ kg/m^3 floats above water with density $\rho_w = 1000$ kg/m^3. A solid cylinder is floating so that initially 1/3 of its height is in the water, 1/3 is in the oil, and 1/3 is in the air. Additional oil is added until the cylinder is floating *only in oil*. What fraction of the cylinder's height is then submerged in the oil?

1. 3/5
2. 3/4
3. 2/3
4. 8/9
5. 4/5

Problem 3

A block floats partially submerged in a container of liquid. The entire container is then accelerated *upward*. Assume both the liquid and the block are incompressible. Which of the following happens?

1. The block descends lower into the liquid.
2. The block ascends higher in the liquid.
3. The block does not ascend nor descend in the liquid.
4. The answer depends on the direction of motion of the container.
5. The answer depends on the rate of change of the acceleration.

Problem 4

A warlord wants to know the weight of a live elephant but has only a small scale and a boat. You may use the small scale *and* the boat (on a calm lake) but nothing else. **Using Archimedes' principle**, design a procedure to determine the elephant's weight. State precisely what you would measure with the small scale, what marks you would make on the boat, and how you would combine those measurements to obtain the elephant's weight. No numerical values are required.

9 Quick Reference Values

Table 1: Common densities (near room temperature).

Water (liquid)	$\rho \approx 1000\,\text{kg/m}^3$
Sea water	$\rho \approx 1025\,\text{kg/m}^3$
Ice	$\rho \approx 920\,\text{kg/m}^3$
Vegetable oil	$\rho \approx 800\,\text{kg/m}^3$ to $900\,\text{kg/m}^3$
Air	$\rho \approx 1.2\,\text{kg/m}^3$
Mercury	$\rho \approx 13\,600\,\text{kg/m}^3$

10 Additional Resources

- **MIT 8.01 Lecture 28:** Hydrostatics, Archimedes' Principle, and Fluid Dynamics.
- **MIT 8.01 Lecture 27:** Gases and Incompressible Liquids.